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SUBJECT CODE NO: - RNP-8008
FACULTY OF SCIENCE AND TECHNOLOGY
F.E. (Group A) Mech/ Civil (NEP) Sem - I
Examination June/July 2025
Engineering Mechanics

[Time: 3:00 Hours]

[Max. Marks: 60]

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q.nos 2, 3, 4, 5, 6 and 7

SECTION A

Q1 Answer the following questions.

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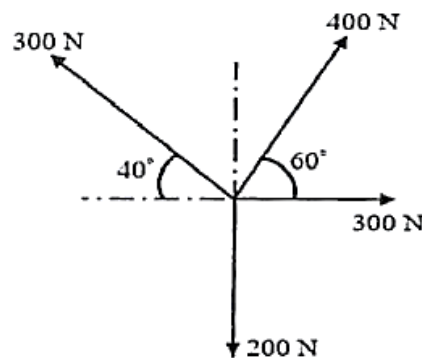
1. The result of two or more forces acting on a body is called:
 - a) Moment
 - b) Resultant force
 - c) Torque
 - d) Equilibrium
2. Which of the following is a non-coplanar force system?
 - a) All forces lie in the same plane
 - b) Forces are at different angles in space
 - c) All forces act parallel to one another
 - d) All forces have a common point of action
3. What is the key characteristic of a couple in a force system?
 - a) Two equal forces acting in the same direction
 - b) Two forces of different magnitudes acting in the same direction
 - c) Two equal, opposite forces separated by a distance
 - d) Two forces acting at right angles to each other
4. Which method is used to analyze forces in a truss?
 - a) Direct integration method
 - b) Method of Joint
 - c) Moment distribution method
 - d) Newton's method
5. In a plane truss, if a member is under tension, the force in the member is:
 - a) Positive
 - b) Negative
 - c) Zero
 - d) Undefined
6. What is the centroid of a plane surface?
 - a) The point where the area is maximum
 - b) The point where the first moment of the area is zero
 - c) The point of intersection of the axes of symmetry
 - d) The geometric center of the area
7. Which of the following shapes has its centroid at the geometric center?
 - a) Triangle
 - b) Parabola
 - c) Rectangle
 - d) Semicircle

8. Which of the following is true about a system in equilibrium?
- The sum of all forces and the sum of all moments must be zero.
 - Only the sum of all forces must be zero.
 - Only the sum of all moments must be zero.
 - None of the above.
9. The center of mass of an object is:
- The point where the total weight of the object acts.
 - The point where all the mass of the object is concentrated.
 - The point where all forces acting on the object balance each other.
 - The point where the object has maximum density.
10. Which law of motion states that for every action, there is an equal and opposite reaction?
- Newton's First Law
 - Newton's Second Law
 - Newton's Third Law
 - Law of Universal Gravitation

SECTION B

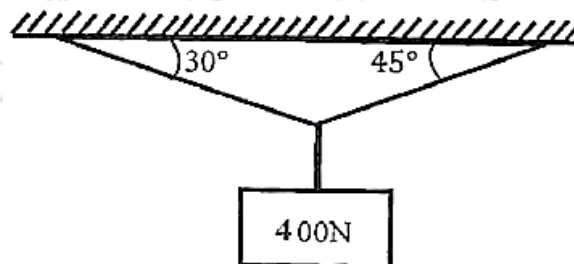
Q2 a) Find magnitude and direction of given concurrent force system.

05



b) A weight of 400N is attached by two strings calculate the tension in the string

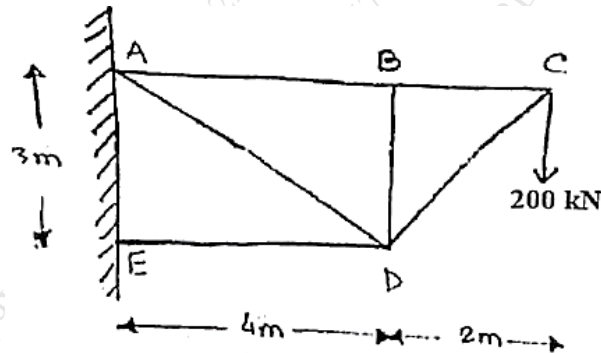
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Q3

Find out the magnitude of forces in given truss.

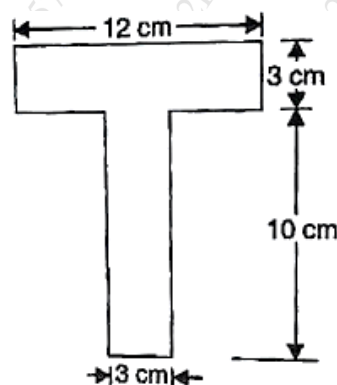
10



Q4

Determine the centroid of T-section shown in figure below about X and Y axis.

10



Q5

Displacement equation of a particle is given by,

10

$$S = 25t + 6t^2 - 2t^3$$

Find:

- Velocity and acceleration at start
- Time when particle reaches its maximum velocity
- The maximum velocity of the particle.

Q6

- a) A ball is thrown with a velocity of 50 m/s at an angle of elevation of 40° with the horizontal. 05

Find:

- Maximum height reached by the ball
- Time of flight
- Range of ball

- b) A bullet is fired from a gun with an initial velocity of 250 m/s to hit a target. 05
The target is located at a horizontal distance of 3500 m and 650 m above the gun. Determine minimum angle of projection so that the bullet will hit the target.

Q7

An automobile of mass 1.2 tons moves down a 30° incline at a speed of 50 kmph, when the brakes are applied causing a constant braking force of 5 kN. Determine the distance travelled by the Automobile as it comes to rest. 10